ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

NATURAL SCIENCES

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 8 Natural Sciences paper 1 · practice d.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples. Penalise missing units once per question.

SECTION A

QUESTION 1

1 1.1 C ✓ 1.2 B ✓ 1.3 A ✓ 1.4 C ✓ 1.5 A ✓ 1.6 D ✓ 1.7 B ✓ 1.8 C ✓ 1.9 D ✓ 1.10 A ✓ (10)

QUESTION 2

2.1 True ✓✓ (2)

2.2 False ✓ – sink ✓ (2)

2.3 True ✓✓ (2)

2.4 False ✓ – straight ✓ (2)

2.5 True ✓✓ (2)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3

3.1 Photosynthesis: in the green parts / chloroplasts of plants ✓; takes in carbon dioxide ✓. Respiration: in all living cells (mitochondria) ✓; takes in oxygen ✓. (4)

3.2 Any TWO ✓✓: energy for respiration; stored as starch; used to make cellulose for cell walls; used to make proteins and oils for growth; stored in fruit and seeds (2)

3.3.1 Jackal ✓ (1)

3.3.2 There is more food (rabbits) for the jackals ✓, so more jackals survive and have young ✓. (2)

3.3.3 The larger number of jackals eat more rabbits ✓, so the rabbits are killed faster than they can reproduce ✓. (2)

3.4 They break down dead plants, animals and waste ✓ and return nutrients to the soil so plants can use them again (and stop dead material from piling up) ✓. (2)

3.5 Any THREE ✓✓✓: moisture / water; warmth (a suitable temperature); food (nutrients); oxygen (for most); time; a suitable pH (3)

3.6 Any TWO methods ✓✓: refrigerating / freezing; drying; salting; pickling in vinegar; canning; adding sugar (jam); pasteurising; smoking. Explanation of ONE ✓✓, e.g. drying removes the water that micro-organisms need, so they cannot grow; vinegar makes the food too acidic; canning heats the food to kill micro-organisms and seals it so new ones cannot enter. (4)

[20]

QUESTION 4

4.1.1 Diffusion ✓ (1)

4.1.2 The particles of the permanganate and the water are always moving randomly ✓; the purple particles spread from where there are many (near the crystal) to where there are few, until they are evenly mixed ✓. (2)

4.1.3 Faster ✓; in hot water the particles have more energy and move faster ✓. (2)

4.2.1 $\text{Density} = \text{mass} \div \text{volume}$ ✓ $= 54 \div 20$ ✓ $= 2{,}7\ \text{g/cm}^3$ ✓ (3)

4.2.2 It will sink ✓, because its density (2,7 g/cm³) is greater than the density of water ✓. (2)

4.3.1 In the nucleus ✓ (1)

4.3.2 Neutral (no charge) ✓; there are equal numbers of positive protons and negative electrons, so the charges cancel ✓. (2)

4.4 Any TWO metals ✓✓: shiny (lustre); good conductors of heat and electricity; malleable / ductile; mostly solids with high melting points; make a ringing sound (sonorous). Any TWO non-metals ✓✓: dull; poor conductors (insulators); brittle when solid; many are gases at room temperature; low melting points. (4)

4.5.1 Bubbles (fizzing) form and the zinc slowly gets smaller ✓ (1)

4.5.2 Hold a burning splint at the mouth of the test tube ✓; the gas burns with a squeaky "pop" (it is hydrogen) ✓. (2)

[20]

QUESTION 5

5.1 Any TWO ✓✓: length of the wire; thickness (cross-sectional area); type of material; temperature (2)

5.2 (a) Heating effect ✓ (b) Magnetic effect ✓ (c) Heating effect (the filament gets so hot that it glows) / light effect ✓ (3)

5.3 Wind insulated copper wire many times around an iron nail ✓ and connect the ends of the wire to a cell so that current flows ✓. Any ONE way to make it stronger ✓: more turns of wire; more cells (more current); a thicker iron core. (3)

5.4 All the current that flows to the appliance must flow through the fuse ✓, so if the current becomes too high the fuse melts and breaks the whole circuit, protecting the appliance and wiring ✓. (2)

5.5 Torch, ball and wall drawn in a line ✓; straight rays from the torch passing the edges of the ball ✓; dark shadow on the wall between the rays, labelled ✓ (3)

5.6 White light contains all the colours ✓; a white shirt reflects all the colours ✓, so it looks white. A black shirt absorbs all the colours ✓ and reflects almost no light, so it looks black ✓. (4)

5.7 Lightning and thunder happen at the same time ✓, but light travels very much faster than sound ✓, so the light reaches us almost immediately while the sound takes longer ✓. (3)

[20]

QUESTION 6

6.1.1 Venus ✓ (1)

6.1.2 Venus has a very thick atmosphere of carbon dioxide ✓ that traps heat (a strong greenhouse effect) ✓; Mercury has almost no atmosphere, so its heat escapes. (2)

6.1.3 In general, the further a planet is from the Sun ✓, the lower its average temperature ✓ (Venus is the exception). (2)

6.2 SALT (the Southern African Large Telescope) ✓ (1)

6.3 An optical telescope collects visible light (with mirrors or lenses) ✓; a radio telescope collects radio waves with large dishes or antennas, and can also be used in the day and through clouds ✓. (2)

6.4 Light from the star takes 4 years to reach us ✓, so we see the star as it was 4 years ago ✓. (2)

6.5 The Milky Way is a spiral galaxy ✓, a flat disc with curved arms around a bulge in the centre ✓; our solar system is in one of the outer arms, far from the centre ✓. (3)

6.6 Find the Southern Cross and the two Pointer stars ✓; extend the long axis of the cross about four and a half times its length ✓; drop a line straight down from that point to the horizon – that direction is south ✓. (3)

6.7 The Moon does not make its own light; we see the part lit by the Sun ✓, and as the Moon orbits the Earth we see different amounts of its sunlit half ✓. (2)

6.8 A natural satellite ✓ (1)

6.9 Any ONE ✓: it has no atmosphere to burn up meteoroids before they hit; it has no wind or water to wear the craters away. (1)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE